Branching

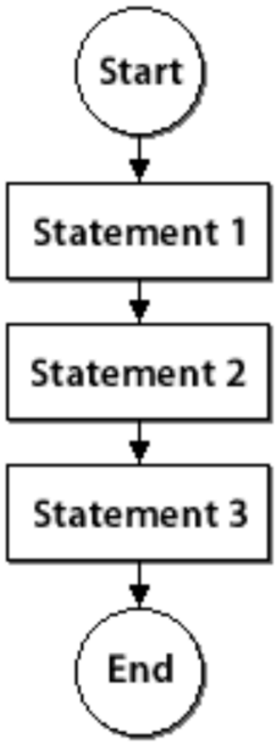
Introduction to computer programming

Management and Artificial Intelligence

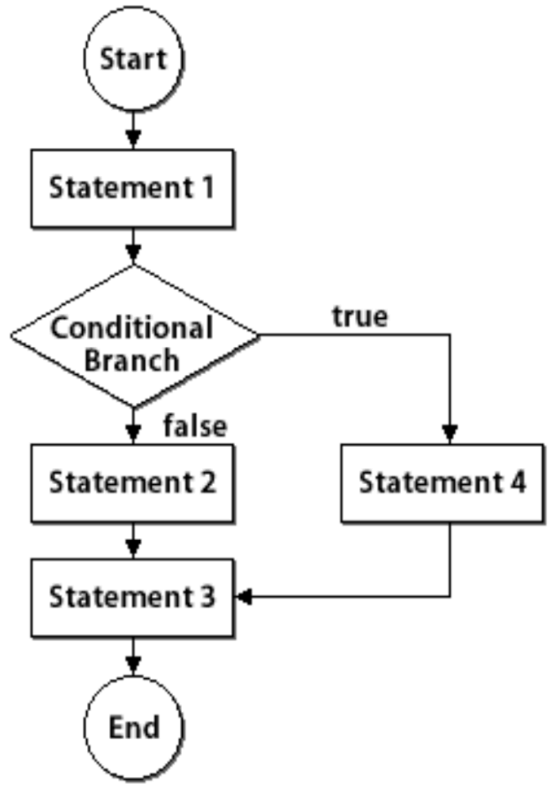
LUISS

Branching

A program without branching



A program with branching



Branching in Python

Plain if

```
if <condition>:
    statement
    statement
    ...
```

if-else

```
if <condition>:
    statement
    ...
else:
    statement
    ...
```

if-elif-else

```
if <condition>:
    statement
    ...
elif <condition>:
    statement
    ...
else:
    statement
    ...
```

- `<condition>` yields `True` or `False`
- Evaluate statements in that block if `<condition>` is `True`
- Indentation is the key to delimit blocks
- Place `:` at the end of `<condition>`

Examples: if

In [16]:

```
num = 3
if num > 0:
    print(num, "is a positive number.")
print("This is always printed. It's outside of the block.")
```

```
3 is a positive number.
This is always printed. It's outside of the block.
```

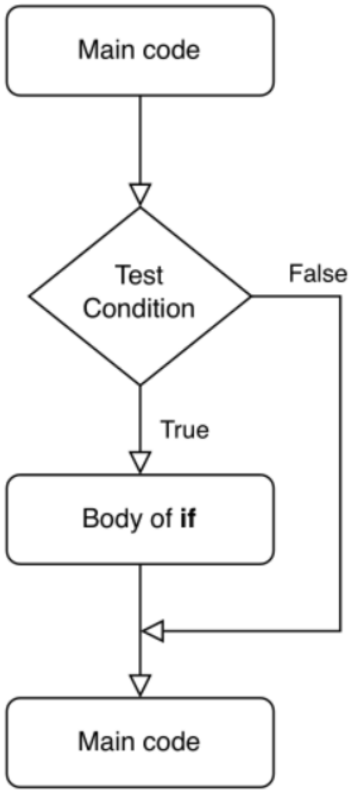
In [17]:

```
num = -1
if num > 0:
    print(num, "is a positive number.")
print("This is always printed. It's outside of the block.")
```

```
This is always printed. It's outside of the block.
```

Flowchart of an `if` statement

Logical flowchart of an if statement:



Examples: if-else

In [18]:
```
num = 1
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

```
Number is positive or zero.
```

What happens with:

- `num = 0`
- `num = -5`

?

In [19]:
```python
num = 0
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

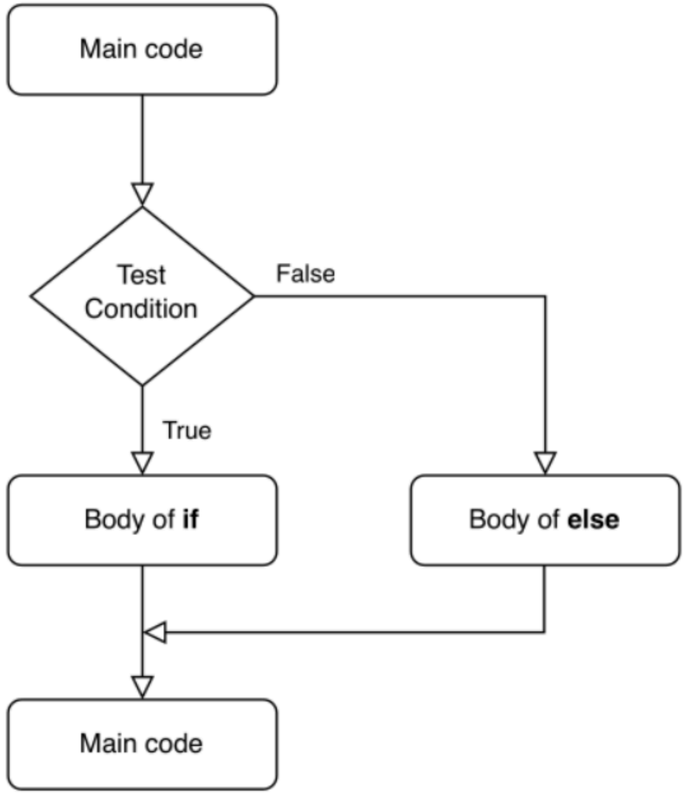
```
Number is positive or zero.
```

In [20]:
```python
num = -5
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

```
Negative number.
```

Flowchart of an `if-else` statement

Logical flowchart of an if-else statement.:



Examples: if-elif-else

In [21]:

```
num = 3.4
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

```
Number is positive.
```

What happens with:

- `num = 0`
- `num = -4.5`

?

In [22]:
```
num = 0
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

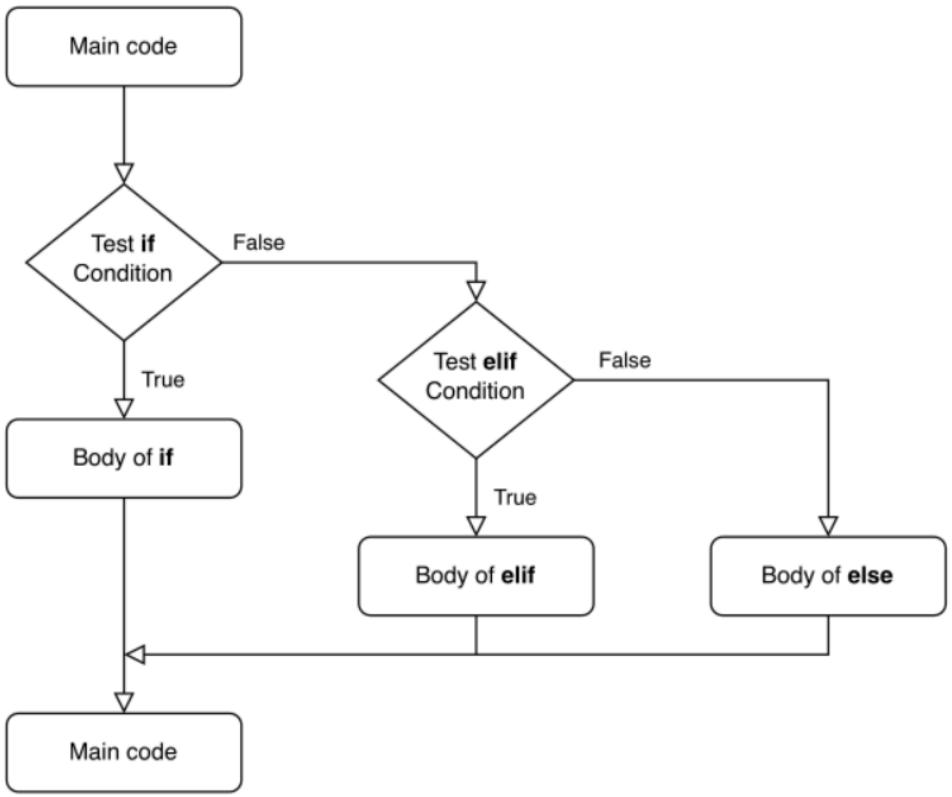
```
Number is zero.
```

In [23]:
```
num = -4.5
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

```
Number is negative.
```

Flowchart of an `if-elif-else` statement

Logical flowchart of an if-elif-else statement:



Nested Statements

Branching statements `if`, `else` and `elif` can be nested arbitrarily.

In [26]:

```
num = input()
num = float(num) # recall: input() returns a string
print("You entered: ", num)
if num >= 0:
    if num == 0:
        print("Number is zero")
    else:
        print("Number is positive")
else:
    print("Number is negative")
```

```
You entered:  10.0
Number is positive
```

Indentation

Recall: Indentation (tab or 4 spaces) delimits blocks of code in Python.

In [25]:

```
x = float(input("Enter a value for x: "))
y = float(input("Enter a value for y: "))
print("x is", x)
print("y is", y)
if x == y:
    print("y and x are equal")
    if y != 0:
        print("Finally, x/y is", x / y)
elif x > y:
    print("y is smaller")
else:
    print("x is smaller")
```

```
x is 10.0
y is 20.0
x is smaller
```

Exercise 1

Write a piece of code that checks whether a number is even. If so, print a message.

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Write a piece of code that checks whether a number is even. If so, print a message.

A possible solution follows:

In [27]:

```
num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
```

```
Condition evaluates to True.
Number 42 is even
```

Exercise 2

Write a piece of code that checks whether a number is even. Print a message about the outcome (odd or even).

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Write a piece of code that checks whether a number is even. Print a message about the outcome (odd or even).

A possible solution follows:

In [28]:
```python
num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
else:
    print("Condition evaluates to False.")
    print("Number", num, "is odd")
```

```
Condition evaluates to True.
Number 42 is even
```

Exercise 3

Write a piece of code that checks whether a number is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable, depending on the outcome of the condition. Print the outcome at the end.

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Write a piece of code that checks whether a number is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable, depending on the outcome of the condition. Print the outcome at the end.

A possible solution follows:

In [29]:
```
num = 42
if num % 2 == 0:
    parity = "even"
else:
    parity = "odd"
print("Number", num, "is", parity)
```

```
Number 42 is even
```

Exercise 4

Write a piece of code that checks whether a number is divisible by 2, 3, 5 and 7.

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Write a piece of code that checks whether a number is divisible by 2, 3, 5 and 7.

A possible solution follows:

In [30]:
```
num = 42
if num % 2 == 0:
    print("Number ", num, "is divisible by 2.")
if num % 3 == 0:
    print("Number ", num, "is divisible by 3.")
if num % 5 == 0:
    print("Number ", num, "is divisible by 5.")
if num % 7 == 0:
    print("Number ", num, "is divisible by 7.")
```

```
Number  42 is divisible by 2.
Number  42 is divisible by 3.
Number  42 is divisible by 7.
```

Exercise 5

Write a piece of code that checks whether a number is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

Exercise 5

Write a piece of code that checks whether a number is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

A possible solution follows:

In [31]:

```python
num = 15
if num % 2 == 0:
    print("Number", num, "is even.")
    if num % 3 == 0:
        print("Number", num, "is also divisible by 3.")
    else:
        print("Number", num, "is not divisible by 3.")
else:
    print("Number", num, "is odd.")
    if num % 5 == 0:
        print("Number", num, "is also divisible by 5.")
    else:
        print("Number", num, "is not divisible by 5.")
```

```
Number 15 is odd.
Number 15 is also divisible by 5.
```

Exercise 6

Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

Exercise 6

Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

A possible solution follows:

In [33]:
```
h_age = int(input("Input a dog's age in human years: "))
print("Inserted age:", h_age)
if h_age <= 0:
    print("Age must be a positive number.")
else:
    if h_age <= 2:
        d_age = h_age * 10.5
    else:
        d_age = 10.5 + 10.5 + (h_age - 2) * 4
    print("The dog's age in dog's years is:", d_age)
```

```
Inserted age: 20
The dog's age in dog's years is: 93.0
```